CSE 242 – Assignment 1 Write-up

Code: `Assignment1.py` (all numbers and figures below are produced by it). Prepared with AI assistance (Claude); conversation log submitted per course policy.

Part 1: Linear Algebra

Q1 With $A = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$:

$$AB = \begin{pmatrix} 3\cdot 1 + 4\cdot 0 & 3\cdot 2 + 4\cdot(-1) \\ 2\cdot 1 + 1\cdot 0 & 2\cdot 2 + 1\cdot(-1) \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}, \quad BA = \begin{pmatrix} 7 & 6 \\ -2 & -1 \end{pmatrix},$$

$$(A+B)^\top = \begin{pmatrix} 4 & 6 \\ 2 & 0 \end{pmatrix}^\top = \begin{pmatrix} 4 & 2 \\ 6 & 0 \end{pmatrix}, \qquad AB^\top = \begin{pmatrix} 3 & 4 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 11 & -4 \\ 4 & -1 \end{pmatrix},$$

$$\operatorname{tr}(A) = 3 + 1 = 4, \qquad \operatorname{tr}(AB) = 3 + 3 = 6 \ (= \operatorname{tr}(BA) = 7 - 1).$$

Q2 $\det(M - \lambda I) = (2-\lambda)^2 - 1 = 0 \Rightarrow \lambda \in \{1, 3\}$.

- $\lambda = 1$: $(M - I)v = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} v = 0 \Rightarrow v_1 = v_2$, so $v = \frac{1}{\sqrt{2}}(1,1)^\top$. Check: $M(1,1)^\top = (2-1, -1+2)^\top = 1\cdot(1,1)^\top$.
- $\lambda = 3$: $(M - 3I)v = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} v = 0 \Rightarrow v_1 = -v_2$, so $v = \frac{1}{\sqrt{2}}(1,-1)^\top$. Check: $M(1,-1)^\top = (2+1, -1-2)^\top = 3\cdot(1,-1)^\top$.

Q3 Let A be $m \times n$ and B be $n \times m$. Then

$$\operatorname{tr}(AB) = \sum_{i=1}^{m} (AB)_{ii} = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}B_{ji}, \qquad \operatorname{tr}(BA) = \sum_{j=1}^{n} (BA)_{jj} = \sum_{j=1}^{n}\sum_{i=1}^{m} B_{ji}A_{ij}.$$

These are the same finite double sum with the order of summation swapped and scalar multiplication commuting, so $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. ■

Q4 $\det C = 4\cdot 6 - 7\cdot 2 = 10 \neq 0$, so C is invertible. Using $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$,

$$C^{-1} = \frac{1}{10}\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{pmatrix}.$$

Check: $CC^{-1} = \frac{1}{10}\begin{pmatrix} 24-14 & -28+28 \\ 12-12 & -14+24 \end{pmatrix} = I$.

Part 2: Probability, MLE, and MAP

Q1 (a) $\sum_{k\geq 0} 3^k/k! = e^3$, so $c\,e^3 = 1 \Rightarrow \boxed{c = e^{-3}}$ (this is Poisson(3)).
(b) $\mathbb{E}[X] = e^{-3}\sum_{k\geq 1} 3\cdot\frac{3^{k-1}}{(k-1)!} = \boxed{3}$. Also $\mathbb{E}[X(X-1)] = e^{-3}\sum_{k\geq 2} 9\cdot\frac{3^{k-2}}{(k-2)!} = 9$, so $\mathbb{E}[X^2] = 9 + 3 = 12$ and $\boxed{\operatorname{Var}(X) = 12 - 3^2 = 3}$.

Q2 (a) $P(A \cap B) = P(A \mid B)P(B) = 0.6\cdot 0.3 = \boxed{0.18}$. (b) $P(B \mid A) = \dfrac{P(A\cap B)}{P(A)} = \dfrac{0.18}{0.5} = \boxed{0.36}$. (c) $P(A \cup B) = P(A) + P(B) - P(A\cap B) = 0.5 + 0.3 - 0.18 = \boxed{0.62}$.

Q3 Data H,H,T,H,T: $h = 3$ heads in $n = 5$ flips, so $L(\theta) = \theta^3(1-\theta)^2$.

(a) $\log L = 3\log\theta + 2\log(1-\theta)$. Setting $\frac{d}{d\theta}\log L = \frac{3}{\theta} - \frac{2}{1-\theta} = 0$ gives $\boxed{\hat{\theta}_{\text{MLE}} = 3/5}$ (second derivative is negative, so this is a maximum).

(b) Posterior $\propto L(\theta)\cdot 2\theta \propto \theta^4(1-\theta)^2$. Then $\frac{4}{\theta} - \frac{2}{1-\theta} = 0$ gives $\boxed{\hat{\theta}_{\text{MAP}} = 4/6 = 2/3}$. In general it is $(h+1)/(n+1)$; for $h = n$ the maximum is at the boundary $\theta = 1$, which this formula also gives.

Q4 $\log L(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2$.

$$\frac{\partial}{\partial\mu}: \ \frac{1}{\sigma^2}\sum_i (x_i - \mu) = 0 \ \Rightarrow \ \hat{\mu} = \bar{x} = \frac{16}{4} = \boxed{4},$$

$$\frac{\partial}{\partial\sigma^2}: \ -\frac{n}{2\sigma^2} + \frac{\sum_i (x_i-\mu)^2}{2\sigma^4} = 0 \ \Rightarrow \ \hat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \hat{\mu})^2 = \frac{9+1+1+9}{4} = \boxed{5}$$

(divisor n, so this estimator is biased).

Q5 The posterior is $\propto \mathcal{N}(x \mid \mu, 4)\,\mathcal{N}(\mu \mid 0, 1)$. Maximizing the log, $-\frac{(x-\mu)^2}{8} - \frac{\mu^2}{2}$, gives $\frac{x-\mu}{4} - \mu = 0$, so

$$\hat{\mu}_{\text{MAP}} = \frac{x/4}{1/4+1} = \frac{5/4}{5/4} = \boxed{1}.$$

The prior pulls the estimate from $x = 5$ toward 0.

Q6 Let $\mu = \mathbb{E}[X]$. By linearity of expectation (μ is a constant),

$$\text{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2. \ \blacksquare$$

Part 3: Principal Component Analysis

Q1 Let $X \in \mathbb{R}^d$ have mean $\mu = \mathbb{E}[X]$ and covariance matrix

$$\Sigma = \mathbb{E}\big[(X-\mu)(X-\mu)^\top\big], \qquad \Sigma_{ij} = \text{Cov}(X_i, X_j).$$

For a fixed unit vector u, $\mathbb{E}[u^\top X] = u^\top\mu$, so $u^\top X - \mathbb{E}[u^\top X] = u^\top(X-\mu)$. Hence

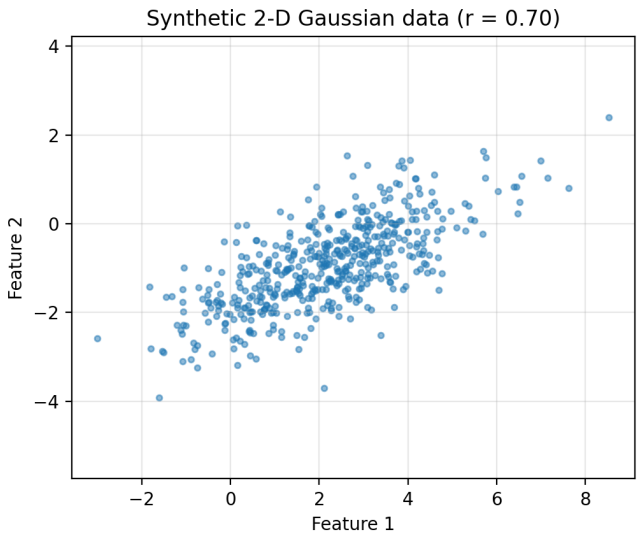
$$\begin{aligned} \text{Var}(u^\top X) &= \mathbb{E}\big[(u^\top(X-\mu))^2\big] = \mathbb{E}\big[u^\top(X-\mu)(X-\mu)^\top u\big] && \text{(a scalar equals its transpose)} \\ &= u^\top\,\mathbb{E}\big[(X-\mu)(X-\mu)^\top\big]\,u = u^\top\Sigma u. && (u \text{ constant; linearity}) \ \blacksquare \end{aligned}$$

Maximizing $u^\top\Sigma u$ subject to $u^\top u = 1$: the Lagrangian $u^\top\Sigma u - \lambda(u^\top u - 1)$ has gradient $2\Sigma u - 2\lambda u = 0$, i.e. $\Sigma u = \lambda u$. The best direction is the top eigenvector of Σ, and the variance it captures is $u^\top\Sigma u = \lambda$.

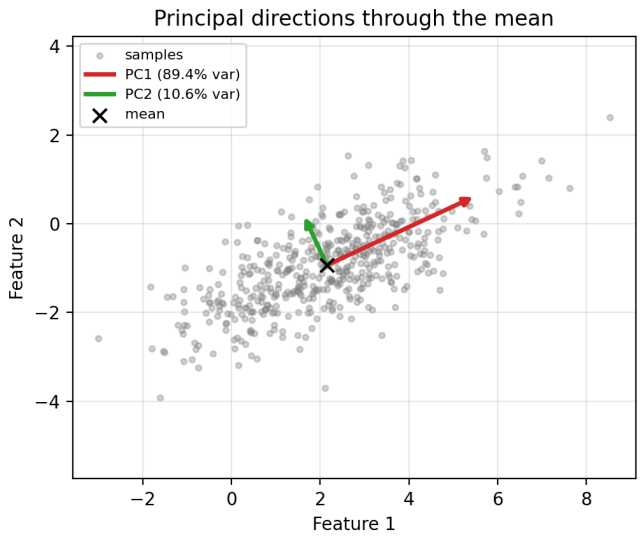
Q2: Synthetic data 500 samples were drawn with NumPy (`default_rng(0).multivariate_normal`) from a bivariate Gaussian with mean $(2, -1)$ and covariance $\begin{pmatrix} 3 & 1.2 \\ 1.2 & 1 \end{pmatrix}$. The true correlation is $1.2/\sqrt{3} \approx 0.69$ and the sample correlation is 0.70. Figure 1 (left) shows a tilted elliptical cloud with positive slope: large Feature 1 values tend to come with large Feature 2 values. Feature 1 has the larger spread (variance 3 vs. 1).

Q3: PCA from scratch Implemented in `pca()` with NumPy only, following the Q1 derivation:

1. **Center:** $X_c = X - \bar{x}$, the sample analogue of $X - \mu$.
2. **Covariance:** $S = X_c^\top X_c/(n-1)$, the sample analogue of Σ.
3. **Eigendecompose:** $S = V\Lambda V^\top$ via `eigh` (S is symmetric, so eigenvalues are real and eigenvectors orthonormal). By the Lagrange argument in Q1, each eigenvector maximizes $u^\top Su$ among directions orthogonal to the earlier ones, with variance λ.
4. **Select:** sort eigenvalues in descending order and keep the top eigenvectors as principal components; scores $= X_c V$.



(a) Q2: synthetic data.



(b) Q3: principal directions through the mean (arrow length $= 2\sqrt{\lambda}$).

Figure 1: Data and PCA directions.

Results: $\bar{x} = (2.15, -0.93)$, $S = \begin{pmatrix} 2.92 & 1.17 \\ 1.17 & 0.95 \end{pmatrix}$, $\boxed{\lambda_1 = 3.459,\ \lambda_2 = 0.411}$, PC1 $\approx (0.91, 0.42)$ (up to sign), PC2 $\approx (-0.42, 0.91)$. In Figure 1 (right), PC1 runs along the long axis of the cloud (maximum variance) and PC2 is perpendicular to it. The eigenvalues are close to those of the true covariance (3.56, 0.44), up to sampling error.

Q4: Explained variance The fraction for component i is $\lambda_i / \sum_j \lambda_j$, where the total variance is $\mathrm{tr}(S) = 3.869$:

$$\boxed{\text{PC1: } 89.4\%, \qquad \text{PC2: } 10.6\%} \quad \text{(Figure 2)}.$$

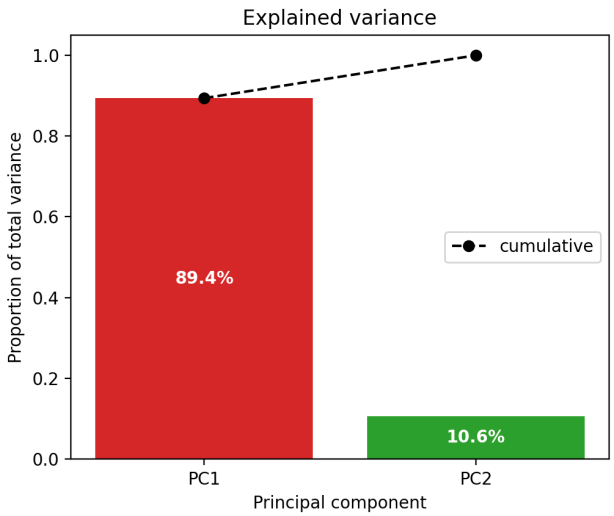


Figure 2: Explained variance fractions and cumulative curve (Q4).

Discussion. Because the features are strongly correlated, one direction captures almost 90% of the variance, so projecting from 2-D to 1-D along PC1 loses only about 10.6%. Dimensionality reduction cuts storage, computation, and noise and makes data easier to visualize, but every dropped component discards its share of variance. The trade-off is usually managed by keeping enough components to reach a target cumulative variance (e.g. 90–95%) or by looking for an "elbow" in the scree plot. If the features were uncorrelated with equal variance, no direction would dominate and little could be dropped for free.